



Relationship between coupling constants in Heisenberg exchange Hamiltonian and Ising model



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ABSTRACT

We show that the coupling constant involved in the Heisenberg exchange Hamiltonian is related to the coupling constant calculated in the Ising model by a simple expression of the form $J(\text{Hei}) = J(\text{Ising})/N$ where N is the average number of equivalent magnetic sites per cell taken for calculation. This relation is demonstrated by DFT calculations on the crystallographic geometry of MnSb alloy in different magnetic phases, and on the optimized geometries of polymers of meta-xylylene and the silicon substituted counterparts.

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1. Introduction

It is well-known that the Heisenberg exchange Hamiltonian serves as the effective spin Hamiltonian for ferromagnetic (FM) and antiferromagnetic (AFM) solids, though in practice the Hamiltonian in Ising approximation is often used for calculations and to obtain qualitative theoretical guidance [1–3]. It is desirable to have a clearly defined relationship between the coupling constants in Heisenberg Hamiltonian and those calculated from Ising Hamiltonian, and whenever possible, compare these with the coupling constants that can be estimated from experiment. This question is addressed here. A relationship is obtained, and it is illustrated by exemplary calculations on three systems.

2. Theoretical background

The treatment of ferromagnetism in solids in the absence of an external magnetic field is generally based on the Heisenberg effective spin Hamiltonian,

$$H_{\text{Hei}} = E_0 - 2 \sum_j \sum_{\varepsilon} J_{\varepsilon}^H \mathbf{S}_j \cdot \mathbf{S}_{j+\hat{n}_{\varepsilon}}. \quad (1)$$

In the above J_{ε}^H is the exchange coupling constant in Heisenberg model between the neighbouring magnetic sites joined by the unit vector $\hat{n}_{\varepsilon}$ along the crystal axis ε , and $\mathbf{S}_j$ is the operator for the spin angular momentum at site j . The Ising model, often applied to

metallic or alloy systems [4], relies on a special case of Heisenberg spin exchange in Eq. (1), with all spins considered as directed along a specific (z) axis:

$$H_{\text{Ising}} = E_0^{\text{Ising}} - 2 \sum_j \sum_{\varepsilon} J_{\varepsilon}^I (S_j)_z (S_{j+\hat{n}_{\varepsilon}})_z. \quad (2)$$

Here J_{ε}^I is the magnetic exchange coupling constant in Ising model between neighbouring sites along crystal axis ε , and $(S_j)_z$ is the operator for the z -component of spin angular momentum. The coupling constant J_{ε}^I involves the interaction only between the z -components of spin, and therefore, differs from J_{ε}^H that is used in the general exchange Hamiltonian with $\mathbf{S}_i \cdot \mathbf{S}_j$ term. Ising model has often been used for periodic systems to investigate qualitative trends in properties [4].

2.1. Coupling constants

Let us write the number of sites as $N_{\text{site}} = N_{\text{uc}} N_{\text{mag}}$ where N_{uc} is the number of unit cells per unit volume and N_{mag} is the number of 'equivalent' magnetic sites per unit cell. The 'equivalent' magnetic sites are sites of same type of atoms in same chemical environment and carrying the same spin, but they may appear at different relative topological positions in the repeating unit as illustrated by Figure 1.

The total spin operators are given by

$$\mathbf{S}_{\text{total}} = \sum_j \mathbf{S}_j, \quad S_{\text{tot},z} = \sum_j (S_j)_z. \quad (3)$$

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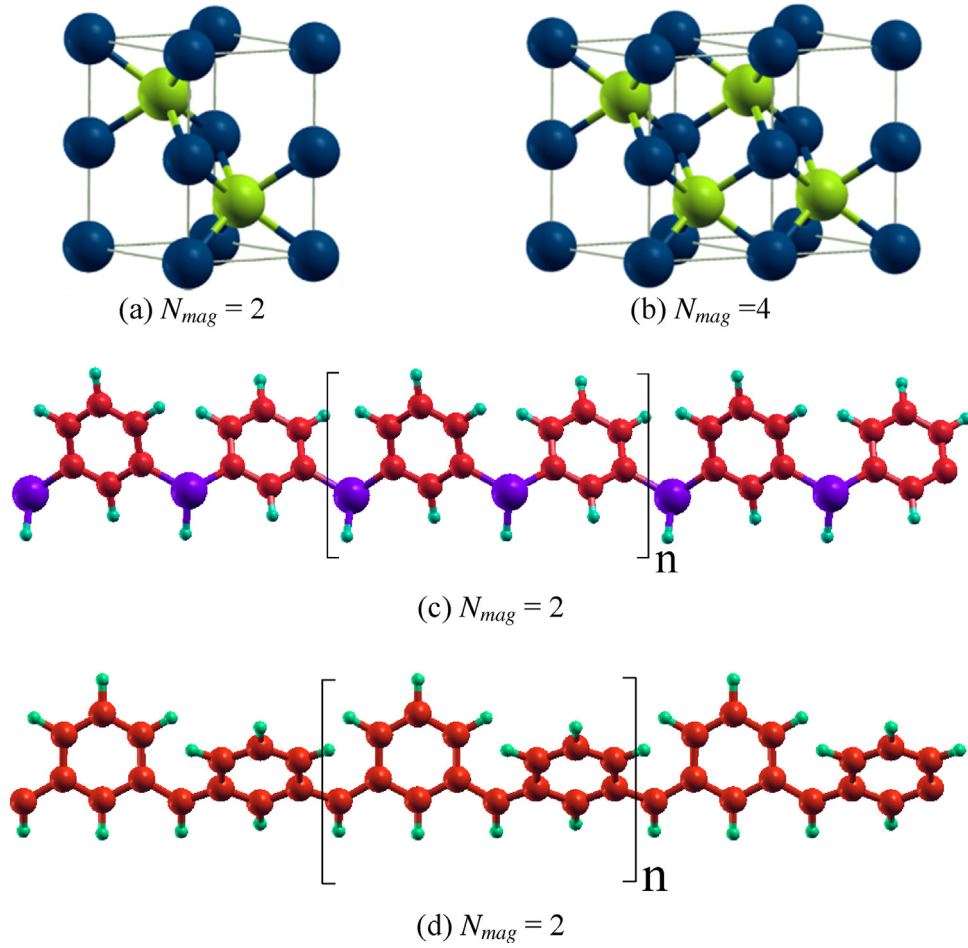


Figure 1. (a) Elementary unit cell for MnSb ($N_{mag} = 2$); (b) the supercell for MnSb ($N_{mag} = 4$); (c) unit cell for silicon substituted 1-D meta-xylylene polymer ($N_{mag} = 2$) and (d) unit cell for 1-D meta-xylylene polymer ($N_{mag} = 2$). Atoms are coloured as follows: Mn (dark blue), Sb (light green), C (red), Si (purple), and H (green).

The spin state vectors $|\{m_i\}\rangle$ where m_i varies over all possible eigenvalues of $(S_i)_z$ for every site i form a complete basis set for the common eigenstates of S_{total}^2 and $S_{tot,z}$. We note

$$S_{tot,z}|\{m_i\}\rangle = M|\{m_i\}\rangle, \quad M = \sum_j^{\text{site}} m_j \quad (4)$$

and

$$\langle\{m_i\}|S_{total}^2|\{m_i\}\rangle = S_{\max}(S+1) + M^2 - \sum_j^{\text{site}} m_j^2, \quad S_{\max} = N_{\text{site}}S. \quad (5)$$

For the FM state, $m_j = S$ for every site j so that $M = S_{\max} = N_{\text{site}}S$, and $\langle S_{total}^2 \rangle = S_{\max}(S_{\max} + 1)$. This also yields the spin per cell $S_{\max}/N_{uc} = N_{mag}S$ so that $\langle S^2 \rangle_{\text{cell}} = N_{mag}S(N_{mag}S + 1)$. For any AFM configuration, however, $M = 0$ that can correspond to any S_{total} ($S_{total} = 0, 2S, 4S, \dots, S_{\max}$). For any particular AFM arrangement with m_j alternatively varying as S and $-S$ along a particular crystal axis ε' while maintaining FM arrangement along all other crystal axes, $\sum_j m_j^2 = S_{\max}S$ so that $\langle S_{total}^2 \rangle = S_{\max}$, and similarly $\langle S^2 \rangle_{\text{cell}} = N_{mag}S$. These results are exact.

The total (space-spin) electronic Hamiltonian H_{el} can be used to calculate the energy of any state with a particular spin configuration under Born–Oppenheimer approximation. This energy can be equated to the energy of a spin Hamiltonian. As there are $\mathbf{z}_\varepsilon$

nearest neighbours along each direction ε , we get from the Ising Hamiltonian

$$E_{\text{FM}} = E_0^{\text{Ising}} - N_{\text{site}}S^2 \sum_{\varepsilon}^{\text{axis}} \mathbf{z}_\varepsilon J_\varepsilon^I, \quad (6)$$

$$E_{\text{AFM}}^{\varepsilon'} = E_{\text{FM}} + 2N_{\text{site}}S^2 \mathbf{z}_{\varepsilon'} J_{\varepsilon'}^I.$$

where ε is a running index and the AFM spin arrangement is present along the specific axis ε' .

The energy per cell is written as $E_c = E/N_{uc}$ such that

$$E_{c,\text{FM}} = E_{c,0}^{\text{Ising}} - N_{mag}S^2 \sum_{\varepsilon}^{\text{axis}} \mathbf{z}_\varepsilon J_\varepsilon^I, \quad (7)$$

$$E_{c,\text{AFM}}^{\varepsilon'} = E_{c,\text{FM}} + 2N_{mag}S^2 \mathbf{z}_{\varepsilon'} J_{\varepsilon'}^I.$$

This gives

$$2\mathbf{z}_\varepsilon J_\varepsilon^I = \frac{E_{c,\text{AFM}}^{\varepsilon'} - E_{c,\text{FM}}}{N_{mag}S^2} \quad (8)$$

for each crystal axis ε .

There arises a problem in the calculation of the Heisenberg coupling constant J_ε^H . The spin state $M = S_{\max}$ is necessarily a pure spin state representing $S_{total} = S_{\max}$, that is, the FM ground state configuration is indeed the exact ground state configuration of the FM solid. As far as an explicit quantum mechanical calculation is concerned, the calculated solution for $M = 0$ is in general a mixture of states with different values of S_{total} as observed earlier after Eq. (5). Anderson showed that the true AFM ground state in any solid is a harmonic

oscillator state involving different sublattices. Spins of the two sublattices are in opposite directions. One cannot define the spin direction of any sublattice on an average basis [5]. As a specific spin arrangement cannot represent the AFM ground state, the Heisenberg coupling constant cannot be directly calculated from it. Klein et al. have discussed the limitations of the spin wave approach [6].

2.2. Broken symmetry

Any unrestricted self-consistent-field (SCF) calculation (Hartree–Fock or Kohn–Sham) on a specific AFM configuration in Ising model is in principle comparable to the ‘broken symmetry’ calculations in quantum chemistry on the less-than-highest-spin states of a polyradical. Two issues are involved here. First, there is selective localization of down spin on specific sites [7]. The resulting solution is an eigenstate of S_z , but not of S_{total}^2 . Second, the problem is more intrinsic. For example, one needs a two-determinant configuration to describe an open-shell singlet diradical. The unrestricted procedure gives a solution for $S_z = 0$ with both spatial and spin symmetries broken and $\langle S^2 \rangle \approx 1$. To calculate the singlet–triplet energy gap for a diradical, the DFT broken symmetry (BS) methodology was developed by Noodleman [8,9].

The Heisenberg spin Hamiltonian for a diradical

$$H = -2J^H S_1 \cdot S_2 \quad (9)$$

gives a singlet–triplet energy difference of $2J^H$. For an intramolecular ferromagnetic interaction, $J^H > 0$, and for an antiferromagnetic coupling, $J^H < 0$. Noodleman derived the relation [8,9]

$$J^N = \frac{E_{\text{BS}} - E_T}{1 + S_{ab}^2} \quad (10)$$

where S_{ab} is the overlap integral between the two magnetically active orbitals a and b , and E_{BS} and E_T are the energies of the broken symmetry state and the triplet state, respectively. The coupling constant J^N gives a fair estimate of J^H in the two-electron-in-two-orbital model when S_{ab} is small. Real molecules, however, deviate from the two-electron-two-orbital model, and for them the spin polarization treatment led to two different formulae valid in the limits of weak [10,11] and strong [12,13] overlap. For a general system, however, a single expression was formulated by Yamaguchi et al. [14–16],

$$J^Y = \frac{(E_{\text{BS}} - E_{\text{HS}})}{\langle S^2 \rangle_{\text{HS}} - \langle S^2 \rangle_{\text{BS}}} \quad (11)$$

where E_{BS} and E_{HS} are energies, and $\langle S^2 \rangle_{\text{BS}}$ and $\langle S^2 \rangle_{\text{HS}}$ are the expectation values of the square of total spin angular momentum operator, for broken-symmetry (BS) and high-spin (HS) solutions, respectively. The J^Y is an estimate of J^H . At each limit of S_{ab} , the Yamaguchi expression reduces to the corresponding limiting form.

Considering the $\langle S^2 \rangle$ values per cell, that is, $\langle S^2 \rangle_{\text{FM}} = N_{\text{mag}} S(N_{\text{mag}} S + 1)$ and $\langle S^2 \rangle_{\text{AFM}} = N_{\text{mag}} S$, one obtains

$$2z_{\varepsilon} J_{\varepsilon}^Y = \frac{E_{\text{c,AFM}}^{\varepsilon} - E_{\text{c,FM}}^{\varepsilon}}{N_{\text{mag}}^2 S^2} \quad (12)$$

As J^Y is an estimate of the Heisenberg coupling constant, the latter is related to the normally calculated Ising model coupling constant in (8) by

$$J_{\varepsilon}^H = \frac{J_{\varepsilon}^I}{N_{\text{mag}}} \quad (13)$$

It equals the usually calculated coupling constant in Ising model only when $N_{\text{mag}} = 1$.

Though the two models are about 90 years old, relationship (13) remained largely unnoticed because of two reasons. First, due to

limitations of computational resources, serious quantum mechanical calculations on the magnetic properties of periodic systems have been done within the last two decades. For instance, Dovesi et al. reported UHF calculations on KMF_3 ($M = \text{Mn, Fe, Co and Ni}$) in 1997 [17]. Second, almost all reported calculations have been made on the ‘single site model’, that is, the repeating units have been selected such that $N_{\text{mag}} = 1$. A relationship between the coupling constants was not explicitly sought. Towler et al. calculated the Ising exchange parameters for MnO and NiO using single site model in 1994 and obtained the AFM–FM energy difference that led to a good estimate of Néel temperature [18]. Jacobson et al. directly calculated the Heisenberg coupling constants for the same systems in 2013 by considering a progressively large cell in real space, and found good agreement with a variety of experimental results [19].

Experimental results such as the ferromagnetic transition temperature (T_C) and saturation magnetization can be correlated with the Heisenberg coupling constant. The molecular field theory gives the ratio

$$\frac{k_B T_C}{z \bar{J}} = \frac{2S(S+1)}{3} \quad (14)$$

that equals 0.5 for $S = 1/2$. This expression is often used by experimentalists. Here z is the total number of nearest neighbours, and $\bar{J}$ is the average of coupling constants of the Heisenberg Hamiltonian. Also, the EPR g factor (spectroscopic splitting factor) is taken to be 2. Somewhat better statistical treatments by Rushbrooke and Wood give the ratio 0.28 for simple cubic lattice and a slightly larger value for fcc and bcc lattices [20]. If we retain the value 0.28 for monoclinic cells, we get

$$\bar{J} = \frac{5.36 k_B T_C}{2zS(S+1)} \quad (15)$$

Replacing gS ($=2S$) by n_B , the number of ferromagnetic magnons per site, the average coupling constant can be estimated from the ferromagnetic transition temperature by

$$\bar{J}_{\text{est}} = \frac{10.72 k_B T_C}{zn_B(n_B + 1)} \quad (16)$$

This expression gives a standpoint for comparing the calculated numbers with the observed transition temperature and saturation magnetic moment that provides n_B .

3. Test cases

There have been excellent discussions on the magnetic properties of inorganic solids such as KMF_3 systems (with $M = \text{Mn, Fe, Co, and Ni}$) [17], MnO and NiO [18,19]. The studies in Refs. [17,18] involved $N_{\text{mag}} = 1$, while in Ref. [19] the convergence of the Heisenberg exchange parameters were explicitly studied for different cut-offs in real space.

In this work we investigate three systems, all having two equivalent magnetic sites per repeating unit. The first one is MnSb, a purely inorganic solid [Figure 1(a) and (b)]. The second one is Si-substituted poly-meta-xylylene with silicon atoms as radical sites [Figure 1(c)]. The third is a purely organic polymer of meta-xylylene [Figure 1(d)].

3.1. Alloy MnSb

The alloy MnSb has NiAs type structure (space group $P6_3/mmc$) and hcp lattice with $a = 4.285 \text{ \AA}$ and $c = 6.113 \text{ \AA}$ [21]. For MnSb, n_B is known ($n_B = 3.5$) [22]. There are in average six Mn atoms and six Sb atoms per unit cell. For computational purposes one can utilize the elementary unit cell that is shown in Figure 2. The elementary unit cell has in average two MnSb pairs. Three phases, namely, ferromagnetic (FM), antiferromagnetic along z -axis (AFM1) and

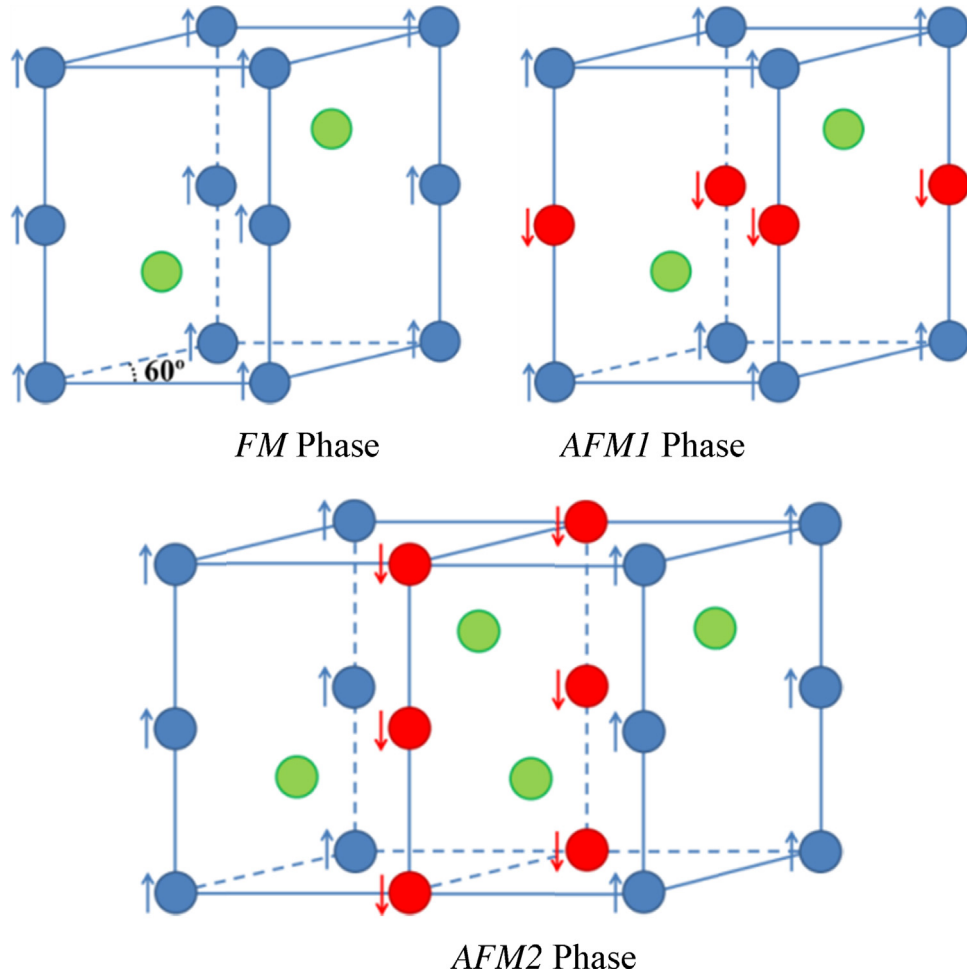


Figure 2. Elementary unit cell of hexagonal symmetry, showing different magnetic phases FM and AFM1, and the supercell showing AFM2: blue circles denote spin up Mn, red circles are for spin down Mn, and green circles stand for Sb. Each elementary cell contains in average 2 Mn atoms and 2 Sb atoms, and the corresponding supercell contains 4 atoms of each type. This supercell is generated using the expansion matrix 200/010/001.

antiferromagnetic in xy plane (AFM2) are illustrated in the same figure.

For calculations on periodic systems, the energy of FM and AFM1 phases can be obtained by taking the elementary unit cell illustrated in Figure 2 as the repeating unit. However, the energy of the AFM2 phase has to be determined by considering the supercell containing four Mn atoms as the repeating unit, and the energies of FM and AFM1 phases can be recalculated for this supercell. The supercell energy differences are to be divided by 2 to obtain the energy differences *per* elementary unit cell.

From the Ising Hamiltonian in Eq. (2) we get the following values for total energy *per* elementary cell

$$\begin{aligned}
 E_{c,FM} &= E_{c,0}^{\text{Ising}} - N_{\text{mag}} S_z^2 [\mathbf{z}_z J_z^I + \mathbf{z}_{xy} J_{xy}^I] \\
 E_{c,AFM1} &= E_{c,0}^{\text{Ising}} - N_{\text{mag}} S_z^2 [-\mathbf{z}_z J_z^I + \mathbf{z}_{xy} J_{xy}^I] \\
 E_{c,AFM2} &= E_{c,0}^{\text{Ising}} - N_{\text{mag}} S_z^2 \left[\mathbf{z}_z J_z^I - \frac{\mathbf{z}_{xy}}{3} J_{xy}^I \right]
 \end{aligned} \quad (17)$$

where $N_{\text{mag}}=2$ for the elementary unit cell, and $\mathbf{z}_z$ and $\mathbf{z}_{xy}$ are the numbers of nearest neighbour atoms along z -axis and in the xy plane, respectively, for each magnetic site. For hcp lattice, $\mathbf{z}_z=2$ and $\mathbf{z}_{xy}=6$. Furthermore, J_z^I and J_{xy}^I are the corresponding

nearest-neighbour Ising coupling constants. One obtains the relations

$$2\mathbf{z}_z J_z^I = \frac{(E_{c,AFM1} - E_{c,FM})}{N_{\text{mag}} S_z^2}, \quad (18)$$

$$\frac{4}{3}\mathbf{z}_{xy} J_{xy}^I = \frac{(E_{c,AFM2} - E_{c,FM})}{N_{\text{mag}} S_z^2}. \quad (19)$$

For MnSb in FM, AFM1 and AFM2 phases, $S=|S_z|=2$ for each Mn atom. When we take the same periodicity into account, that is, explicitly consider the difference in the number of nearest-neighbour antiferromagnetic interactions in the two involved phases, (number of antiferromagnetic interactions in AFM1 or AFM2 phase plus number of ferromagnetic interactions in FM phase), we must write

$$\begin{aligned}
 2\mathbf{z}_z J_z^Y &= \frac{(E_{c,AFM1} - E_{c,FM})}{N_{\text{mag}}^2 S_z^2}, \\
 \frac{4}{3}\mathbf{z}_{xy} J_{xy}^Y &= \frac{(E_{c,AFM2} - E_{c,FM})}{N_{\text{mag}}^2 S_z^2}.
 \end{aligned} \quad (20)$$

Furthermore,

$$\begin{aligned}
 \bar{J}_{\text{Ising}} &= \frac{1}{4}(J_z^I + 3J_{xy}^I), \\
 \bar{J}_{\text{Hei}} &= \bar{J}^Y = \frac{\bar{J}_{\text{Ising}}}{2}.
 \end{aligned} \quad (21)$$

These are the working equations for the system under study.

Table 1

Single point energies calculated for FM and AFM1 phases of the elementary unit cell from the crystallographic geometry of the MnSb alloy. All energy values are in atomic units.

	$E_{c,FM}$ (S_z per MnX)	$E_{c,AFM1}$ (S_z per MnX)	$\Delta E_{c(AF1-FM)}$	J_z^I (cm^{-1}) Eq. (15)
MnSb				
B3LYP	−218.80951 (2.0070)	−218.79621 (0.0017)	0.01331	91.3
B3PW	−219.10288 (2.0101)	−219.08845 (0.0015)	0.01442	98.9
PBE0	−218.94131 (2.0146)	−218.92396 (0.0009)	0.01734	118.9

Table 2

Single point energies calculated for FM, AFM1 and AFM2 phases of the supercell from the crystallographic geometry of alloy MnSb. All energy values are in atomic units.

	Phase	E (S_z per MnX)	$\Delta E_{(AFM1-FM)}$	$\Delta E_{(AFM2-FM)}$	J_z^I (cm^{-1})	J_{xy}^I (cm^{-1})	$\bar{J}_{\text{Ising}}$ (cm^{-1})
B3LYP	FM	−437.61893 (2.0075)	0.02716	0.02331	93.1	39.9	53.2
	AFM1	−437.59177 (−0.0083)					
	AFM2	−437.59562 (0.0000)					
B3PW	FM	−438.20478 (2.0107)	0.02926	0.02523	100.4	43.3	57.6
	AFM1	−438.17552 (−0.0017)					
	AFM2	−438.17955 (0.0000)					
PBE0	FM	−437.88252 (2.0143)	0.03464	0.02863	118.8	49.1	66.5
	AFM1	−437.84788 (−0.0009)					
	AFM2	−437.85389 (0.0000)					

The *ab-initio* solid state computational code CRYSTAL09 [23,24] is used for energy calculations by DFT methodology. Experimental lattice constants from Ref. [21] are adopted as input. Hybrid functionals are known to provide better results for insulators and semiconductors [25–28]. A possible problem with applicability exists only for conductive materials. This has been discussed by Franchini [29]. The ‘time-tested’ B3LYP functional generates relative energies close to the experimental values and good coupling constants [30–33], and B3PW reproduces J near experimental value [33,34]. PBE0 is known to produce band structure data comparable to experiments [35].

We employ the pseudopotential basis sets for the atomic valence states, namely (i) 3s, 3p, 4s and 3d orbitals of Mn, and (ii) 5s and 5p orbitals of Sb: Mn_HAYWSC-411d311_heifets.2005 [33], and Sb_DURAND-21d1G_causa.1991 [36]. The chosen Mn basis set with B3LYP functional has produced good estimates of structural parameters and magnetic exchange coupling constants for LaMnO_3 [33] and ScMnO_3 [37]. The basis set selected for Sb generally leads to structural parameters close to the experimental ones [36]. Use of the pseudopotentials allows the size of basis set to be reduced and thereby save computational resources.

Total energy, (S_z), and the charge and spin populations are determined in the following way. Single point calculations are done on the elementary unit cells and the corresponding supercells. SPINLOCK keyword is used to maintain the difference between the number of electrons with spins up and down to get the desired spin states (such as $S_z = 0, 1, \dots, 8$). However, the spin state with $S_z = 4$ for the elementary unit cell and that with $S_z = 8$ for the supercell come out to be most stable. To get the second antiferromagnetic phase (AFM2) and the in-plane coupling constant, an expansion matrix of 100/020/001 is used to generate the supercell. The spin state 8 is employed for the ferromagnetic phase of this supercell. Shrinking factor of 12 is used to obtain eigenvectors and eigenvalues. Tolerance parameters (TOLINTEG) have five integers T1 T2 T3 T4 T5 with default values 6 6 6 6 12 in CRYSTAL09. To get greater accuracy in the coulomb and exchange integrals, tolerance parameters 7 7 7 7 15 are used as recommended in Refs. [38,39]. Convergence is achieved with the help of ANDERSON procedure [40]. The Fock-KS matrix mixing (FMIXING) (for two successive cycles) is taken as 50%. For all the calculations, the starting electronic configurations have been taken as $4s^2 3d^5$ for Mn and $5s^2 5p^3$ for Sb.

Table 3

Average magnetic exchange coupling constant (in cm^{-1}) calculated from the experimental transition temperature and n_B of MnSb solid.

	Rushbrooke–Wood coupling constant		
	$\bar{J}_{\text{Ising}}^a$	$\bar{J}_{\text{Hei}}^b$	$\bar{J}_{\text{est}}^c$
B3LYP	53.2	26.6	28.4
B3PW	57.6	28.8	
PBE0	66.5	33.2	

^a Ising model calculation, Table 1.

^b Corrected for Heisenberg exchange by using $N_{\text{mag}} = 2$ in Eq. (13).

^c Heisenberg model, mean field theory and statistical approximation, Eq. (16).

3.2. Polyradicals

The calculations on meta-xylylene polyradical chains and the corresponding silicon substituted species are done using code GAUSSIAN 09 [41]. Optimization of molecular geometries and the subsequent single point calculations on silicon substituted species are performed using 6-311G(d,p) and 6-311++G(d,p) basis sets, respectively, with B3LYP and B3LYP-D functionals. For meta-xylylene polyradical chains the molecular geometry optimization and subsequent single point calculations are done using 6-31G(d,p) and 6-311G(d,p) basis sets, respectively, using B3LYP and M06-2X functionals.

CRYSTAL09 code is used for calculations on the infinitely long one-dimensional polymers. The geometry of the unit cell is fully optimized in both FM and AFM states while using B3LYP and B3LYP-D functionals with TZVP basis sets for Si, C and H atoms [42]. The M06-2X functional is not available in CRYSTAL09. For poly-meta-xylylene and its silicon substituted form, we have used (i) SHRINK factors 16 16 and 32 32, respectively, (ii) the SPINLOCK keyword, and (iii) tolerance parameters (TOLINTEG) 7 7 7 7 14. Convergence is achieved using LEVSHIFT option 1 0 along with 30% FMIXING.

4. Results and discussion

Single point calculations are done on the experimentally reported crystal geometries. The input files and selected outputs are included in Supporting Information. The energy differences and calculated Ising model coupling constants are given in Table 1 (elementary unit cell) and Table 2 (supercell). The Ising model coupling

Table 4

Calculated magnetic exchange coupling constants in cm^{-1} : The number of phenylene rings in the polymer is p . The J_{∞} is estimated by assuming an exponential behaviour. Periodic calculations are done by using B3LYP and B3LYP-D functionals with TZVP basis set.

Si substituted poly- <i>meta</i> -xylylene			Poly- <i>meta</i> -xylylene		
p	B3LYP/6-311++G(d,p)	B3LYP-D/6-311++G(d,p)	p	B3LYP/6-311G(d,p)	M06-2X/6-311G(d,p)
1	243.0	243.0	2	1682	1542
2	178.4	178.4	3	572.8	521.3
3	166.5	169.7	4	464.7	353.3
4	122.1	122.1	∞	453.0	320.2
∞	76.7	85.6			

Periodic calculation	B3LYP/TZVP	B3LYP-D/TZVP	Periodic calculation ^a	B3LYP/TZVP
J^I	187.1	197.0	J^I	1077
J^H	93.6	98.5	J^H	538.5

Data have been taken from Ref. [45].

^a M06-2X is not included in CRYSTAL09 code.

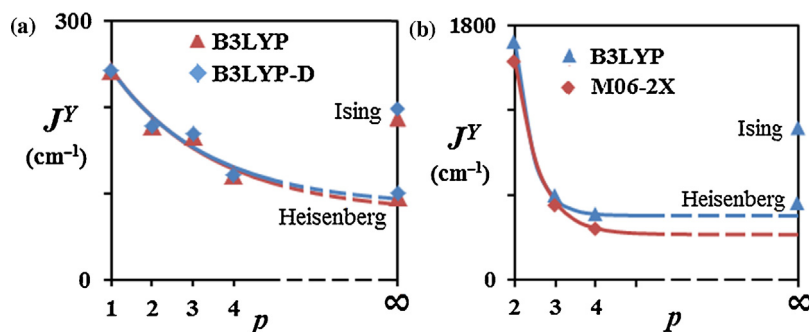


Figure 3. Exponential fitting of decreasing J values with increase in chain length for (a) silicon substituted poly-*meta*-xylylene (rmsd 9.23 for B3LYP and 10.49 B3LYP-D) and (b) poly-*meta*-xylylene (rmsd 45.5 for B3LYP and 0.16 M06-2X). The J value reaches a limiting value J_{∞} that compares with J^H . The periodic calculation gives J^I and relation (13) using $N_{\text{mag}} = 2$ leads to J^H .

constant along the c -axis (z -axis) that is obtained from the supercell calculation is almost equal to that from the calculation on the elementary unit cell. The in-plane coupling constant J_{xy}^I is obtained from Eq. (19). It is also positive, confirming that MnSb is most stable in the FM phase. The average $\bar{J}_{\text{Ising}}$ is obtained from Eq. (21).

As the elementary unit cell in Figure 1 resembles the combination of two monoclinic cells, each having one MnSb pair in average, we adopt the Rushbrooke–Wood expression (15) for this alloy. The ferromagnetic transition temperature T_C is 587 K for MnSb [43] so that $\bar{J}_{\text{est}} = 28.4 \text{ cm}^{-1}$ in the Heisenberg model. In Table 3, coupling constants from crystal geometries are compared with the coupling constant estimated from n_B and T_C . The calculated Heisenberg coupling constants are in striking agreement with the estimated value from Eq. (15), and vindicate the relationship between $\bar{J}_{\text{Ising}}$ and $\bar{J}_{\text{Hei}}$ in Eq. (21).

We have also optimized the geometry of the elementary unit cell and the supercell. The detailed results are not included in this report and will be published elsewhere.

In meta-xylylene polyradicals and the silicon substituted forms, the high-spin state is always the ground state. A decreasing trend in J values with length is observed in both cases. This trend is shown in Table 4. Matsuda et al. have shown that the decay of J in phenylene chains is generally exponential in form [44]. An exponential decay is illustrated in Figure 3 where a limiting value (J_{∞}) is found from the calculated J^Y values. We get J_{∞} (in cm^{-1}) = 76.7 (B3LYP), 85.6 (B3LYP-D) for the silicon substituted polymer, and 453.0 (B3LYP), 320.2 (M06-2X) for the unsubstituted meta-xylylene polymer.

From periodic calculations we obtain J^I (in cm^{-1}) = 187.1 (B3LYP), 197.0 (B3LYP-D) for the silicon polymer, and 1077 (B3LYP), 1096 (B3LYP-D) for the unsubstituted meta-xylylene polymer. Using Eq. (13), we get J^H (in cm^{-1}) = 93.5 (B3LYP), 98.5 (B3LYP-D) for the silicon polymer, and 538.5 (B3LYP), 548 (B3LYP-D) for the

unsubstituted meta-xylylene polymer. Thus the limiting values (J_{∞}) estimated from the Heisenberg coupling constants in Yamaguchi expression are more or less equal to the J^H values and not the J^I ones. The J^H is exactly half of the J^I that is directly obtained from the results of periodic calculations. A somewhat negative deviation of J_{∞} from J^H is evident. With increasing number of phenylene rings p , the geometry of the polyradical progressively deviates from the periodicity that is obtained by repeating the dimer unit. The detailed results will be published elsewhere [45].

Because of the difference in composition, B3PW is expected to yield the best J value, followed by B3LYP, (the former functional containing an improved correlation part), and then the adjustable-parameter-free PBE0. This has been precisely observed in our calculation of Heisenberg coupling constant for MnSb. B3LYP is considered as time-tested for thermochemical calculations. For organic molecules, M06-2X has a greater contribution from HF exchange, and generally produces a better (and smaller) coupling constant than B3LYP does. For polyradicals, the effect of the dispersive forces can be quite significant, as found from the periodic calculations on the silicon-substituted polymer.

5. Conclusions

This study gives an insight into the relationship between the coupling constant of Heisenberg exchange and that normally calculated from Ising approximation. We find relation (13) and demonstrate it by sample calculations on MnSb. The two sets of coupling constants are equal in the ‘single site model.’ Thus the choice of the repeating unit becomes critical. This is also valid for polyradicals. When two successive radical monomers are twisted such that they are alternately periodic, the dimer is the repeating unit and a sensible result evolves only after dividing Ising model

coupling constant by 2. These and related issues will be discussed in our future work.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.cplett.2015.01.001](https://doi.org/10.1016/j.cplett.2015.01.001).

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